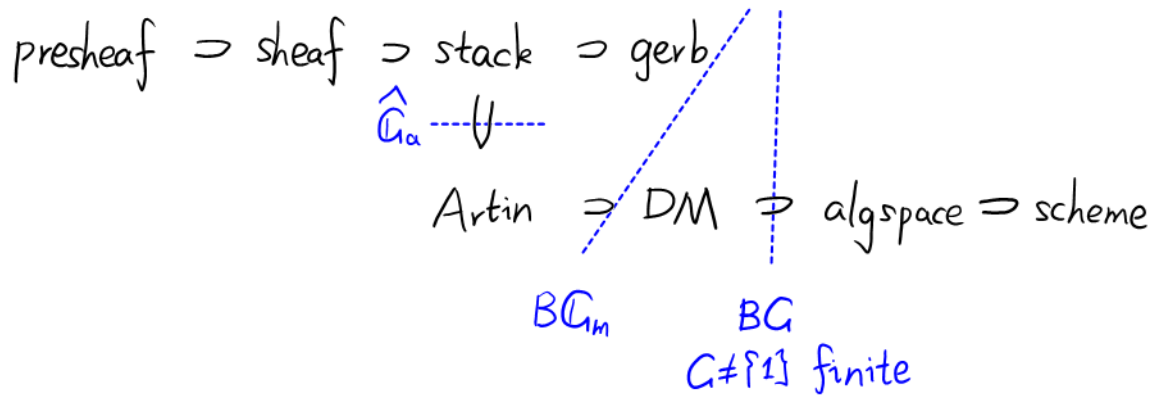


Eine Woche, ein Beispiel
 5.18 gerb & twisted sheaves

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$$\begin{aligned}
 \hat{G}_a &= \text{Spf } k[[t]] && \longrightarrow && \text{Spec } k[[t]] \\
 &= \varinjlim_n \text{Spec } k[t]/(t^n) && && \\
 &= \{ (t) \} && && = \{ (0), (t) \}
 \end{aligned}$$

Def. (stack in groupoids) Let $\mathcal{F} \in Sh(X)$.

$\mathcal{F}: Open(X)^{\text{op}} \longrightarrow \text{Gpd}$ is a stack, if

$$\mathcal{F}(\mathcal{U}) \xrightarrow{\sim} \text{holim} \left(\prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_{ij}) \rightrightarrows \prod_{i,j,k} \mathcal{F}(U_{ijk}) \right)$$

as an equivalence of groupoids, where

$$\begin{aligned} Ob(\text{holim}(\dots)) &= \{ \text{descent datum } (\mathcal{L}_i, \phi_{ij}) \} \\ &= \left\{ (\mathcal{L}_i, \phi_{ij})_{i,j} \mid \begin{array}{l} \mathcal{L}_i \in \mathcal{F}(U_i) \\ \phi_{ij}: \mathcal{L}_i|_{U_{ij}} \xrightarrow{\sim} \mathcal{L}_j|_{U_{ij}} \\ \phi_{jk} \circ \phi_{ij}|_{U_{ijk}} = \phi_{ik}|_{U_{ijk}} \end{array} \right\} \end{aligned}$$

$$\text{Mor}((\mathcal{L}_i, \phi_{ij}), (\mathcal{L}'_i, \phi'_{ij})) = \left\{ (f_i)_i \mid \begin{array}{ccc} f_i: \mathcal{L}_i & \xrightarrow{\sim} & \mathcal{L}'_i \\ \mathcal{L}_i|_{U_{ij}} & \xrightarrow{f_i|_{U_{ij}}} & \mathcal{L}'_i|_{U_{ij}} \\ \phi_{ij} \downarrow & \circlearrowright & \downarrow \phi'_{ij} \\ \mathcal{L}_j|_{U_{ij}} & \xrightarrow{f_j|_{U_{ij}}} & \mathcal{L}'_j|_{U_{ij}} \end{array} \right\}$$

E.g. $Sh^{\sim} \supset Q\text{Coh}^{\sim} \supset \text{Coh}^{\sim} \supset B\mathbb{G}_m$
 stack of sheaves
 are stacks, where

$$\begin{aligned} Ob(Sh^{\sim}(\mathcal{U})) &= \{ \text{sheaves on } \mathcal{U} \} & = Ob(Sh(\mathcal{U})) \\ \text{Mor}_{Sh^{\sim}(\mathcal{U})}(\mathcal{F}, \mathcal{F}') &= \{ \text{Iso of sheaves on } \mathcal{U} \} & \subseteq \text{Mor}_{Sh(\mathcal{U})}(\mathcal{F}, \mathcal{F}') \end{aligned}$$

$$\begin{aligned} Ob(B\mathbb{G}_m(\mathcal{U})) &= \text{Pic}(\mathcal{U}) \\ \text{Mor}_{B\mathbb{G}_m(\mathcal{U})}(\mathcal{L}, \mathcal{L}') &= \text{Isom}_{\mathcal{O}_{\mathcal{U}}}(\mathcal{L}, \mathcal{L}') \end{aligned}$$

e.g. $\text{Mor}_{B\mathbb{G}_m(\mathcal{U})}(\mathcal{L}, \mathcal{L}) = \mathbb{G}_m(\mathcal{U})$

Def (Gerby) A stack $\mathcal{G}: \text{Open}(X) \rightarrow \text{Gpd}$ is called a gerby, if (i) + (ii) + compatibility:

- (i) $\forall \mathcal{U}$, $\exists$ an open cover $\mathcal{U} = \bigcup_i \mathcal{U}_i$ s.t. $\mathcal{G}(\mathcal{U}_i) \neq \emptyset$;
(ii) $\forall \mathcal{L}, \mathcal{L}' \in \mathcal{G}(\mathcal{U})$, $\exists$ an open cover $\mathcal{U} = \bigcup_i \mathcal{U}_i$ s.t. $\mathcal{L}|_{\mathcal{U}_i} \cong \mathcal{L}'|_{\mathcal{U}_i}$.

Furthermore, for G : sheaf of gps on X , we call a stack $\mathcal{G}$ as a G -gerby, if (i) + (ii)' + compatibility:

- (ii)' $\forall \mathcal{L} \in \mathcal{G}(\mathcal{U})$, $\exists$ an open cover $\mathcal{U} = \bigcup_i \mathcal{U}_i$ s.t. $\mathcal{L}|_{\mathcal{U}_i} \cong B(G|_{\mathcal{U}_i})$;

or: $\forall \mathcal{L} \in \mathcal{G}(\mathcal{U})$, $\exists$ an open cover $\mathcal{U} = \bigcup_i \mathcal{U}_i$ s.t. $\text{Aut}_{G(\mathcal{U})}(\mathcal{L}) \cong G|_{\mathcal{U}_i}$
 $\uparrow$
sheaf of automorphisms

E.g. BG is a G -gerby over X .
 ${}^k\mathcal{I}$ is a μ_k -gerby over X , ${}^k\sqrt{\mathcal{O}_X} \cong B\mu_k$.

Here,

$$\begin{aligned} G &: \text{sheaf of gps on } X \\ \text{Ob}(BG(\mathcal{U})) &= \{G|_{\mathcal{U}}\text{-torsors on } \mathcal{U}\} \\ \text{Mor}(\mathcal{L}, \mathcal{L}') &= \{\varphi: \mathcal{L} \xrightarrow{\sim} \mathcal{L}' \text{ is } G|_{\mathcal{U}}\text{-equivalent}\} \\ \text{e.p. } \text{Mor}(G|_{\mathcal{U}}, G|_{\mathcal{U}}) &= G(\mathcal{U}) \end{aligned}$$

$$\begin{aligned} \text{Ob}({}^k\mathcal{I}(\mathcal{U})) &= \left\{ (M, \varphi) \mid \begin{array}{l} M \in \text{Pic}(\mathcal{U}) \\ \varphi: M^{\otimes k} \rightarrow \mathcal{L}|_{\mathcal{U}} \end{array} \right\} \\ \text{Mor}((M, \varphi), (M', \varphi')) &= \left\{ \begin{array}{l} \phi: M \rightarrow M' \text{ s.t.} \\ \begin{array}{ccc} M^{\otimes k} & \xrightarrow{\phi^{\otimes k}} & M'^{\otimes k} \\ \varphi \downarrow & \circlearrowleft & \downarrow \varphi' \\ & \mathcal{L}|_{\mathcal{U}} & \end{array} \end{array} \right\} \end{aligned}$$

